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	K.S.RANGASAMYCOLLEGE OF TECHNOLOGY, TIRUCHENGODE – 637 215 (Autonomous)	MATHS
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IMPORTANT QUESTION			
Programme & Branch	B.E. - ECE	Semester	II
Course Code & Name	60 MA 003 & INTEGRALS, PARTIAL DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORM		

S.No	Part – A (Short Answer Questions)	Marks	Blooms level	CO	PO
Unit –I : Multiple Integrals					
1.	Evaluate $\int_0^1 \int_0^2 xy dy dx$	2	Un	1	1
2.	Evaluate $\int_0^2 \int_0^x xy^2 dy dx$	2	Un	1	1
3.	Evaluate $\int_0^3 \int_0^2 e^{x+y} dy dx$	2	Un	1	1
4.	Evaluate $\int_0^3 \int_0^2 xy(x + y) dy dx$	2	Un	1	1
5.	Evaluate $\int \int xy dy dx$ over the region bounded by $x = 0$, $y = 0$ and $x + y \leq 1$	2	Un	1	1
6.	Evaluate $\int_0^{\pi/2} \int_0^{\sin \theta} r dr d\theta$	2	Un	1	1
7.	Evaluate $\int_0^{\pi} \int_0^{\cos \theta} r dr d\theta$	2	Un	1	1
8.	Evaluate $\int_{-\pi/2}^{\pi/2} \int_0^{2\cos \theta} r^2 dr d\theta$	2	Un	1	1
9.	Evaluate $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz$	2	Un	1	1
10.	Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz dx dy dz$	2	Un	1	1
11.	Transform the double integration $\int_0^{\infty} \int_0^{\infty} x^2 dx dy$ into polar co-ordinates	2	Ap	1	1
12.	Change the order of integration $\int_0^1 \int_0^{1-x} f(x, y) dy dx$	2	Ap	1	1

S.No	(Descriptive type questions with two subdivision in each questions)	Part – B	Marks	Blooms level	CO	PO
		Multiple Integrals				
		$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$				
1	i) Find the area enclosed by the ellipse		8	Un	1	1
	ii) Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} x dz dy dx$		7	Un	1	1
	i) Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$		8	Un	1	1
2	ii) Change the double integral from Cartesian coordinates to polar coordinates and then evaluate $\int_0^{2a\sqrt{2ax-x^2}} \int_0^x (x^2 + y^2) dy dx$		7	An	1	1
3	$\int_0^1 \int_{x^2}^{2-x} xy dy dx$ Change the order of integration and hence evaluate		15	Ap	1	1
4	$\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy dy dx$ i) Change the order of integration and hence evaluate Change the double integral from Cartesian coordinates to polar coordinates and		15	Ap	1	1
5	then evaluate $\int_0^{2\sqrt{2x-x^2}} \int_0^x (x^2 + y^2) dy dx$		8	An	1	1
	ii) Find the area between the parabolas $y^2 = 9x$ and $x^2 = 9y$		7	Ap	1	1
	i), Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$		7	Un	1	1
6	ii) Transform the integral into polar co-ordinates and hence evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2 + y^2} dy dx$		8	An	1	1

S.No	Part – A	Marks	Blooms level	CO	PO
	(Short Answer Questions)				
	Unit –II : Vector Calculus				
1.	If $\varphi = 3x^2y - y^3z^2$, find $grad\varphi$ at $(1, -2, -1)$	2	Un	2	1
2.	Find the unit normal vector to the surface $x^2 + 2y^2 + z^2 = 7$ at $(1, -1, 2)$	2	AP	2	1

3.	Find 'a' such that $(2x + y)\vec{i} + (z - ay)\vec{j} + (2az - x)\vec{k}$ is solenoidal.	2	AP	2	1
4.	Prove that $\text{curl}(\text{grad}\phi) = 0$.	2	AP	2	1
5.	State Green's theorem	2	Re	2	1
6.	State Stoke's theorem.	2	Re	2	1
7.	Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is solenoidal	2	AP	2	1
8.	Find the normal derivative of $\phi = xy + yz + zx$ at $(-1, 1, 1)$	2	Un	2	1
9.	State Gauss divergence theorem.	2	Re	2	1
10.	Find $\text{curl } \vec{F}$ where $\vec{F} = x^2y\vec{i} + xz\vec{j} + 2yz\vec{k}$.	2	AP	2	1
11.	Prove that $\text{div } \vec{r} = 3$ and $\text{curl } \vec{r} = 0$	2	AP	2	1
12.	If $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ then prove that $\text{curl } \text{curl } \vec{F} = 0$	2	Un	2	1

Part – B		Marks	Blooms level	CO	PO
S.No	(Descriptive type questions with two subdivision in each questions)				
Vector Calculus					
1	i) Find the angle between the surfaces $x^2 - y^2 - z^2 = 11$ and $xy + yz - zx = 18$ at $(6, 4, 3)$.	8	Ap	2	1
	ii) Find a, b, c if $(x + y + az)\vec{i} + (bx + 2y - z)\vec{j} + (-x + cy + 2z)\vec{k}$ is irrotational	7	Ap	2	1
2	Verify the Gauss divergence theorem for $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$	15	Ap	2	1
3	(i) Show that the vector $\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is both irrotational and solenoidal.	7	Ap	2	1
	ii) Find the values of the constants a, b, c, so that the vector $\vec{F} = (axy + bz^3)\vec{i} + (3x^2 - cz)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational and hence find its scalar potential such that $\vec{F} = \nabla\phi$.	8	Ap	2	1
4	Verify Gauss divergence theorem for $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.	15	Ap	2	1
5	i)Verify Green's theorem in the XY plane for $\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region defined by $x = y^2, y = x^2$.	8	Ap	2	1
	ii) Find the directional derivative of $\phi = x^2 + y^2 + z^2$ at the point $(1, 0, 1)$ in the direction of $2\vec{i} + 3\vec{k}$.	7	Ap	2	1
6	i).Verify Stoke's Theorem for a vector field defined by $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ in the rectangular region in the XOY plane bounded by the lines $x = 0, x = a, y = 0$ and $y = b$.	8	Ap	2	1

ii) Show that the vector $\vec{F} = (y^2 \cos x + z^3)\vec{i} + (2y \sin x - 4)\vec{j} + 3xz^2\vec{k}$ is both irrotational and solenoidal

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S.No	Part – A	Marks	Blooms level	CO	PO
	(Short Answer Questions)				
	Unit –III : Analytic functions and Integrals				
1.	Define analytic function	2	Re	3	1
2.	Show that the function $u = x^2 - y^2 - 2xy - 2x + 3$ is harmonic	2	An	3	1
3.	State any two properties of analytic function.	2	Re	3	1
4.	Test the analyticity of the function $f(z) = \sin z$	2	An	3	1
5.	Find the zero's of $\frac{z^3-1}{z^3+1}$	2	Un	3	1
6.	Find the residue of $f(z) = \frac{z+2}{(z-2)(z+1)^2}$ at $z = -1$	2	Un	3	1
7.	Evaluate : $\int_C \frac{z dz}{z+2}$, where C is the circle $ z =1$ using Cauchy's integral formula	2	Ap	3	1
8.	Evaluate: $\int_C \frac{(z^2 - z + 1)}{(z-1)} dz$, where C is $ z = 2$.	2	Ap	3	1
9.	Find the poles of $f(z) = \frac{z^2}{(z-1)(z-2)^2}$	2	Ap	3	1
10.	State Cauchy's integral formula	2	Re	3	1
11.	Define removable singularity	2	Re	3	1
12.	State Cauchy's integral theorem	2	Re	3	1

		Part – B					
S.No	(Descriptive type questions with two subdivision in each questions)	Marks	Blooms level	CO	PO		
Analytic functions and Integrals							
1	i) Determine the analytic function $f(z)$ whose imaginary part is $v = 3x^2y - y^3 - 6xy + 3y$	8	Ap	3	1		
	ii) If $f(z) = u + iv$ is a regular function of z , then prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) f(z) ^2 = 4 f'(z) ^2$	7	Ap	3	1		
2	i) Show that the function $u = 2x - x^3 + 3xy^2$ is harmonic. Determine its analytic function $f(z)$.	8	Ap	3	1		
	ii) If $u + v = (x - y)(x^2 + 4xy + y^2)$, then find $f(z) = u + iv$ in terms of z	7	Ap	3	1		

3	i) Find the analytic function $f(z) = u + iv$ if	8	Ap	3	1
	$u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$				
4	ii) If the function $f(z) = u + iv$ is analytic, then prove that the family of curves $u(x, y) = c_1$ and $v(x, y) = c_2$ cut orthogonally	7	Ap	3	1
	i) Evaluate: $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $ z = 3$	8	Ap	3	1
5	ii) Evaluate by using Cauchy's residue theorem $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$ where C is the circle $ z = \frac{3}{2}$	7	Ap	3	1
	i) Evaluate: $\int_C \frac{z+4}{z^2+2z+5} dz$, where C is the circle $ z+1-i = 2$ using Cauchy's integral formula.	8	Ap	3	1
6	ii) Evaluate: $\int_C \frac{z dz}{(z-1)(z-2)^2}$, where C is the circle $ z-2 = \frac{1}{2}$	7	Ap	3	1
	i) Evaluate $\int_C \frac{\cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $ z = \frac{3}{2}$ using Cauchy's integral formula.	8	Ap	3	1
	ii) Evaluate $\int_C \frac{1-2z}{z(z-1)(z-2)} dz$ where C is the circle $ z = \frac{3}{2}$ using Cauchy's residue theorem.	7	Ap	3	1

S.No	Part – A (Short Answer Questions)	Marks	Blooms level	CO	PO
	Unit –IV : PDE				
1.	Form the PDE by eliminating the arbitrary constants a and b from $z = (x+a)(y+b)$	2	Un	4	1
2.	Form the PDE by eliminating the arbitrary function f from $z = f(x^2 + y^2)$	2	Un	4	1
3.	Solve $p - q = 0$	2	Ap	4	1
4.	Write the complete solution of $z = px + qy + p^2 + q^2$	2	Un	4	1
5.	Solve : $(D^3 - D^2D' - 8DD'^2 + 12D'^3)z = 0$	2	Ap	4	1
6.	Find the particular integral of $(D^2 + 2DD'^2 + D'^3)z = e^{x-y}$	2	Ap	4	1
7.	Solve the partial differential equation $p + q = x + y$	2	Ap	4	1

8.	Solve $(D^2 - 3DD'^2 + 2D'^3)z = 0$	2	Un	4	1
9.	Form the PDE by eliminating the arbitrary constants a and b from $z = (x^2 + a^2)(y^2 + b^2)$	2	Un	4	1
10.	Solve: $p \tan x + q \tan y = \tan z$	2	Ap	4	1
11.	Write the Lagrange's linear equation and subsidiary equation in partial differential equation.	2	Re	4	1
12.	Solve the partial differential equation $pq = 1$	2	Ap	4	1

S.No	Part – B (Descriptive type questions with two subdivision in each questions)	Marks	Blooms level	CO	PO
PDE					
1	i) Solve the partial differential equation $z = px + qy + p^2 - q^2$	8	Ap	4	1
	ii) Form the PDE by eliminating the arbitrary function f from $z = y^2 + 2g((1/x) + \log y)$	7	Ap	4	1
2	i) Solve the partial differential equation $(D^3 - 7DD'^2 - 6D'^3)z = e^{3x+y} + \sin(x + 2y)$	8	Ap	4	1
	ii) Solve the partial differential equation $(D^2 - 2DD' + D'^2)z = \cos(x - 3y)$	7	Ap	4	1
3	i) Solve the partial differential equation $(D^3 - 3DD'^2 + 2D'^3)z = e^{2x-y} + e^{x+y}$	8	Ap	4	1
	ii) Solve the partial differential equation $p^2 + q^2 + 1 = z$	7	Ap	4	1
4	i) Solve the partial differential equation $(D^2 - DD' - 6D'^2)z = e^{3x+y} + x^2y$	8	Ap	4	1
	ii) Solve the Lagrange's equation $(3z - 4y)p + (4x - 2z)q = 2y - 3x$	7	Ap	4	1
5	i) Solve the partial differential equation $(mz - ny)p + (nx - lz)q = ly - mx$	8	Ap	4	1
	ii) Solve: $z = px + qy + \sqrt{1 - p^2 - q^2}$	7	Ap	4	1
6	i) Solve the partial differential equation $z = px + qy + 2\sqrt{pq}$	8	Ap	4	1
	ii) Solve the Lagrange's equation $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$	7	Ap	4	1

S.No	Part – A (Short Answer Questions)	Marks	Blooms level	CO	PO
Unit –V - Laplace Transform					
1.	State sufficient condition for the existence of Laplace transform of a function.	2	Re	5	1
2.	State Initial value theorem on Laplace transforms.	2	Re	5	1
3.	Find $L[t^3 e^{-2t}]$.	2	Un	5	1
4.	Find $L[e^{3t} \cos 2t]$.	2	Un	5	1
5.	Evaluate $\int_0^{\infty} t e^{-2t} \cos t dt$.	2	Un	5	1
6.	Define Laplace transform.	2	Re	5	1
7.	Find the value of $1 * e^t$	2	Un	5	1
8.	Find $L^{-1}\left[\frac{s}{(s-2)^3}\right]$.	2	Un	5	1
9.	Find $L^{-1}\left[\frac{1}{s-2} + \frac{s}{s^2+1} + \frac{2}{s^2+4} + \frac{1}{s}\right]$	2	Un	5	1
10.	State first shifting theorem.	2	Re	5	1
11.	Find $L^{-1}\left[\frac{1}{s(s+1)^2}\right]$.	2	Un	5	1
12.	State Convolution function on Laplace transforms.	2	Re	5	1

S.No	(Descriptive type questions with two subdivision in each questions)	Laplace Transform	Marks	Blooms level	CO	PO
1.	i) Find the Laplace transform of	$f(t) = \begin{cases} t, & 0 < t < b \\ 2b - t, & b < t < 2b \end{cases}$	8	Ap	5	1
	with $f(t + 2b) = f(t)$					
	ii) Find $L[te^{-t} \cos t]$.		7	Ap	5	1
2.	i) Find the Laplace transform of	$f(t) = \begin{cases} 1, & 0 < t < a \\ -1, & a < t < 2a \end{cases}$	8	Ap	5	1
	with period $2a$					
	ii) Find $L^{-1}\left[\frac{1}{(s+1)(s+2)}\right]$, by using convolution theorem		7	Ap	5	1
3	i) Verify the initial and final value theorem on Laplace transform for the function					
	$f(t) = 1 + e^{-t}(\sin t + \cos t)$		8	Ap	5	1

	ii) Find	$L^{-1}\left[\frac{19s+37}{(s+1)(s-2)(s+3)}\right],$	7	Ap	5	1
	by partial fraction method.					
4.	Solve:	$y''-3y'+2y=2e^{-t},$ given $y(0)=1$ and $y'(0)=1$	15	Ap	5	1
	by using Laplace transform method.					
	i) Find	$L^{-1}\left[\frac{s^2}{(s^2+a^2)^2}\right],$	8	Ap	5	1
5	by using convolution theorem.					
	ii). Find	$L^{-1}\left[\log\left(\frac{s+1}{s-1}\right)\right],$	7	Ap	5	1
6.	Solve:	$y''+4y'+3y=e^{-t},$ given $y(0)=1$ and $y'(0)=0$	15	Ap	5	1
	by using Laplace transform method.					